



Multivariate Tail Conditional Expectation Under a Linear Constraint: A New Risk Measure for Aggregated Exposures

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Abstract:

Most multivariate tail risk measures, such as the multivariate tail conditional expectation (MTCE), assume that all risk factors exceeds their marginal quantiles simultaneously. But, in real word, this assumption sometimes is unrealistic, since systemic events are typically generated by a linear combination of risk factors crossing a critical threshold. So, in this paper, we propose a new risk measure called the multivariate tail conditional expectation under a linear constraint (MTCE-LC). This new risk measure captures the expected risk conditional on a weighted sum of risk factors exceeding a high quantile, so that weakness in one risk factor can be offset by strength in another. This measure can be a good tool for stress testing and capital allocation in systemic risk management. We present an exact expression for the proposed measure under multivariate normal risk factors and also, we carry out a Monte Carlo simulation study to show the accuracy of the theoretical results.

Keywords: Tail conditional expectation, Multivariate risk measure, Linear constraint, Systemic risk, Multivariate normal distribution.

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1 Introduction

As we know, a risk measure is a function from a set of random variables associating the risks to a real number. In this regards, several risk measures have been discussed in the literature, primarily used for determining reserves needed for assets within the financial system. For a single risk factor, one of the most important risk measures is the so-called tail conditional expectation (TCE), which is the expected amount of the worst losses, given the loss exceeds a certain quantile level. Let X be a loss random variable; then, the TCE is defined as

$$TCE_q(X) = E(X | X > x_q), \quad q \in (0, 1),$$

where x_q is a q -th quantile of the distribution of X , which is called the value at risk (VaR) measure. It should be kept in mind that in many applications, the risk managers encounter a set of risks instead of only one risk. For example, in fields such as engineering, finance and insurance, modeling joint risk factors plays a critical role in assessing portfolio risks, pricing correlated assets and designing systems resilient to simultaneous failures. To be specific, suppose the financial institution has p different lines of businesses and $X = (X_1, \dots, X_p)^T$ is a vector of risks, where X_i denotes the risk of the i -th business line at the end of a specified period. In this regards, several works have been carried out to extend the concept of tail risk measures from the univariate case to the multivariate case; one may refer to [Abbasi and Hosseiniard \(2013\)](#); [Landsman et al. \(2016\)](#); [Roozegar et al. \(2025\)](#).

A popular way to extend the TCE measure to a multivariate framework is to consider the total risk of the business lines. For this purpose, let $S = X_1 + \dots + X_p$ denote the total risk of the p business lines. Then, the TCE of the random variable S is defined as

$$TCE_q(S) = E(S | S > s_q).$$

For a pair of risk factors $\mathbf{X} = (X_1, X_2)^T \in [0, \infty)^2$, with $E(X_1) < \infty$, [Acharya et al. \(2010\)](#) extended the definition of TCE to the marginal expected shortfall (MES). This measure is of the form

$$MES_q(X_1, X_2) = E(X_1 | X_2 > x_{2q}).$$

The MES is a vital systemic risk measure utilized for assessing risk contagion between two factors within a system. This measure has found applications in finance and environmental science, as demonstrated by studies of [Caporin and Santucci \(2012\)](#); [Brownlees and Engle \(2015\)](#).

[Landsman et al. \(2018\)](#) proposed a novel multivariate tail conditional expectation (MTCE) under multivariate normal distribution risk factors, which has several advantages. Let $\mathbf{X} \sim N_p(\boldsymbol{\xi}, \boldsymbol{\Omega})$ be a vector of

risks; then, MTCE is defined as

$$MTCE_{\mathbf{x}_q}(\mathbf{X}) = E(\mathbf{X} | \mathbf{X} > \mathbf{x}_q) = E(\mathbf{X} | X_1 > x_{1q_1}, \dots, X_p > x_{pq_p}),$$

where $\mathbf{x}_q$ is a $q \times 1$ vector and x_{iq_i} denotes the q_i -th quantile of the marginal distribution of X_i , for $i \in \{1, \dots, p\}$.

While the classical MTCE provides valuable information about joint tail behavior, it has an important limitation: in many real world applications, risk exposures are not independent, and a systemic risk event is often triggered by a linear combination of risk factors exceeding a critical threshold, rather than by each factor individually exceeding its own quantile. For example, in a bank's portfolio, the total loss $L = c_1X_1 + \dots + c_pX_p$ (with weight c_i representing exposures) may become critical even if some individual risks remain below their quantiles. Similarly, in environmental risk assessment, the overall pollution index — a weighted sum of several pollutants — may exceed a regulatory limit without any single pollutant reaching its extreme quantile. Therefore, in this paper, we introduce a new risk measure, called the multivariate tail conditional expectation under a linear constraint (MTCE-LC). We defined this measure as

$$MTCE_{l_q}^{LC}(\mathbf{X}) = E(\mathbf{X} | L > l_q),$$

where the risk vector $\mathbf{X}$ follows a multivariate normal distribution with mean vector $\boldsymbol{\xi}$ and covariance matrix $\boldsymbol{\Omega}$, $\mathbf{c} = (c_1, \dots, c_p)^T$ is a vector of known weights, and l_q denotes the q -th quantile of the distribution of the linear combination $L = \mathbf{c}^T \mathbf{X}$. This measure captures the expected value of each risk factor conditional on the total linear score exceeding a high threshold. Unlike the MTCE measure, which requires simultaneous exceedances of all marginals, the proposed MTCE-LC allows for compensatory effects between risk factors, making it more flexible and realistic for systemic risk monitoring, capital allocation, and stress testing. Furthermore, when $p = 1$ and $c = 1$, the proposed measure reduces to the univariate TCE, ensuring consistency with previous risk measures.

2 MTCE-LC for multivariate normal risk scenarios

This section focuses on obtaining an exact expression for MTCE-LC within the class of multivariate normal distribution. For this purpose, we need to derive the expectation of the multivariate truncated normal distribution where truncation is based on a single linear combination of the variables, which we denote by MTN-LC (Multivariate Truncated Normal – Linear Combination). We denote this model by $\mathbf{Y} = \mathbf{X} | a < \mathbf{c}^T \mathbf{X} < b$, where $\mathbf{X} \sim N_p(\boldsymbol{\xi}, \boldsymbol{\Omega})$.

Over the last five decades, several works have appeared in the literature discussing the computation of expectations for the truncated multivariate normal distribution; see for example, Tallis (1961); Lee (1979); Horrace (2005); Ho et al. (2012); Roozegar et al. (2020). But, these classical results assume that each coordinate of the multivariate normal vector is truncated independently within a hyper-rectangle. In this section, we consider a fundamentally different scenario where truncation is based on a single linear combination of variables and derive an exact expression for the mean of this distribution. For this purpose, first, we derive the probability density function (pdf) of a univariate component of a standard MTN-LC, which we need for subsequent derivation.

Let c_i , for $i = 1, \dots, p$, denote the i -th component of a p -dimensional vector $\mathbf{c} = (c_1, \dots, c_p)^\top$, and by $\mathbf{c}_{-i}$ the vector obtained from $\mathbf{c}$ by deleting its i -th component. Furthermore, for the standard univariate normal distribution, let $\phi(\cdot)$ and $\Phi(\cdot)$ denote its pdf and cumulative distribution function (cdf), respectively, and $\bar{\Phi}(a, b)$ be the probability in real line (a, b) . Then we have the following lemma.

Lemma 2.1. *Let $Z \sim N_p(\mathbf{0}, \mathbf{I}_p)$, and for $i \in \{1, \dots, p\}$, define $Y_i \stackrel{d}{=} Z_i | (a < \mathbf{c}^T \mathbf{Z} < b)$. Then, the pdf of univariate random variable Y_i can be derived as*

$$f_{Y_i}(y_i) = \frac{\phi(y_i)}{\bar{\Phi}\left(\frac{a}{\sqrt{\mathbf{c}^T \mathbf{c}}}, \frac{b}{\sqrt{\mathbf{c}^T \mathbf{c}}}\right)} \left\{ \Phi(\lambda_i y_i + \tau_i^b) - \Phi(\lambda_i y_i + \tau_i^a) \right\},$$

where

$$\lambda_i = -\frac{c_i}{\sqrt{\mathbf{c}_{-i}^T \mathbf{c}_{-i}}}, \quad \tau_i^a = \frac{a}{\sqrt{\mathbf{c}_{-i}^T \mathbf{c}_{-i}}} \text{ and } \tau_i^b = \frac{b}{\sqrt{\mathbf{c}_{-i}^T \mathbf{c}_{-i}}}.$$

Proof. We can write the random variable Y_i as

$$Y_i \stackrel{d}{=} Z_i | \left(a - c_i Z_i < \mathbf{c}_{-i}^T \mathbf{Z}_{-i} < b - c_i Z_i \right).$$

Then, using Bayes' theorem, we can write the pdf of Y_i as

$$f_{Y_i}(y_i) = \frac{\Pr\left(a - c_i y_i < \mathbf{c}_{-i}^T \mathbf{Z}_{-i} < b - c_i y_i\right) \phi(y_i)}{\Pr(a < \mathbf{c}^T \mathbf{Z} < b)},$$

which yields the desired result. $\square$

Theorem 2.2. *Let $\mathbf{X} \sim N_p(\boldsymbol{\xi}, \boldsymbol{\Omega})$, and define $\mathbf{Y} \stackrel{d}{=} \mathbf{X} | (a < \mathbf{c}^T \mathbf{X} < b)$. Then, the first moments of random vector $\mathbf{Y}$ can be obtained as*

$$E(\mathbf{Y}) = \boldsymbol{\xi} + \mathbf{B}\mathbf{q}(a^*, b^*, \mathbf{c}^*),$$

where $\mathbf{B}\mathbf{B}^T = \boldsymbol{\Omega}$ and the i -th element of the $p \times 1$ vector $\mathbf{q}(a^*, b^*, \mathbf{c}^*)$ is

$$q_i(a^*, b^*, \mathbf{c}^*) = \frac{c_i^*}{\sqrt{\mathbf{c}^{*T} \mathbf{c}^*}} \frac{1}{\bar{\Phi}\left(\frac{a^*}{\sqrt{\mathbf{c}^{*T} \mathbf{c}^*}}, \frac{b^*}{\sqrt{\mathbf{c}^{*T} \mathbf{c}^*}}\right)} \left[\phi\left(\frac{a^*}{\sqrt{\mathbf{c}^{*T} \mathbf{c}^*}}\right) - \phi\left(\frac{b^*}{\sqrt{\mathbf{c}^{*T} \mathbf{c}^*}}\right) \right],$$

with

$$a^* = a - \mathbf{c}^T \boldsymbol{\xi}, \quad b^* = b - \mathbf{c}^T \boldsymbol{\xi}, \quad \text{and} \quad \mathbf{c}^* = \mathbf{B}^T \mathbf{c}. \quad (2.1)$$

Proof. First, let $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I})$, and $\mathbf{Y} \stackrel{d}{=} \mathbf{Z} \mid (a < \mathbf{c}^T \mathbf{Z} < b)$. From the result in Lemma 1, we have

$$E(Y_i) = \frac{1}{\bar{\Phi}\left(\frac{a}{\sqrt{\mathbf{c}^T \mathbf{c}}}, \frac{b}{\sqrt{\mathbf{c}^T \mathbf{c}}}\right)} \int_{-\infty}^{+\infty} y_i \phi(y_i) \left\{ \Phi(\lambda_i y_i + \tau_i^b) - \Phi(\lambda_i y_i + \tau_i^a) \right\} dy_i.$$

Using integration by parts and simplifying, we obtain

$$\begin{aligned} \int_{-\infty}^{+\infty} y_i \phi(y_i) \Phi(\lambda_i y_i + \tau_i^b) dy_i &= \lambda_i \int_{-\infty}^{+\infty} \phi(y_i) \phi(\lambda_i y_i + \tau_i^b) dy_i \\ &= \frac{\lambda_i}{\sqrt{1+\lambda_i^2}} \phi\left(\frac{\tau_i^b}{\sqrt{1+\lambda_i^2}}\right) \int_{-\infty}^{+\infty} \phi\left(y_i; -\frac{\lambda_i \tau_i^b}{1+\lambda_i^2}, \frac{1}{1+\lambda_i^2}\right) dy_i \\ &= -\frac{c_i}{\sqrt{\mathbf{c}^T \mathbf{c}}} \phi\left(\frac{b}{\sqrt{\mathbf{c}^T \mathbf{c}}}\right). \end{aligned}$$

Therefore, in the standard case,

$$E(Y_i) = \frac{c_i}{\sqrt{\mathbf{c}^T \mathbf{c}}} \frac{1}{\bar{\Phi}\left(\frac{a}{\sqrt{\mathbf{c}^T \mathbf{c}}}, \frac{b}{\sqrt{\mathbf{c}^T \mathbf{c}}}\right)} \left[\phi\left(\frac{a}{\sqrt{\mathbf{c}^T \mathbf{c}}}\right) - \phi\left(\frac{b}{\sqrt{\mathbf{c}^T \mathbf{c}}}\right) \right].$$

For the general case, when $\mathbf{X} \sim N_p(\boldsymbol{\xi}, \boldsymbol{\Omega})$, we have $\mathbf{X} \stackrel{d}{=} \boldsymbol{\xi} + \mathbf{B}\mathbf{Z}$, where $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I})$ and $\mathbf{B}\mathbf{B}^T = \boldsymbol{\Omega}$.

So, the random vector $\mathbf{Y} \stackrel{d}{=} \mathbf{X} \mid (a < \mathbf{c}^T \mathbf{X} < b)$ can be written as

$$\begin{aligned} \mathbf{Y} &\stackrel{d}{=} \boldsymbol{\xi} + \mathbf{B}\mathbf{Z} \mid a < \mathbf{c}^T (\boldsymbol{\xi} + \mathbf{B}\mathbf{Z}) < b \\ &\stackrel{d}{=} \boldsymbol{\xi} + \mathbf{B} [\mathbf{Z} \mid a - \mathbf{c}^T \boldsymbol{\xi} < \mathbf{c}^T \mathbf{B}\mathbf{Z} < b - \mathbf{c}^T \boldsymbol{\xi}]. \end{aligned}$$

Letting a^* , b^* and $\mathbf{c}^*$ as in (2.1), we obtain

$$E(\mathbf{Y}) = \boldsymbol{\xi} + \mathbf{B} E\left(\mathbf{Z} \mid a^* < (\mathbf{c}^*)^T \mathbf{Z} < b^*\right),$$

which, using the standard-case result above, gives the expression in the theorem. $\square$

The MTCE-LC expression follows directly from Theorem 1 by letting $a = l_q$ and b tend to infinity. We present this result in the following lemma.

Lemma 2.3. If $\mathbf{X} \sim N_p(\boldsymbol{\xi}, \boldsymbol{\Omega})$, then

$$MTCE_{l_q}^{LC}(\mathbf{X}) = \boldsymbol{\xi} + \mathbf{B}\mathbf{q}(l_q, \mathbf{c}^*)$$

where $\mathbf{c}^*$ is defined as in (2.1) and the i -th element of the $p \times 1$ vector $\mathbf{q}(l_q, \mathbf{c}^*)$ is

$$q_i(l_q, \mathbf{c}^*) = \frac{c_i^*}{\sqrt{\mathbf{c}^{*T} \mathbf{c}^*}} \frac{1}{1 - \Phi\left(\frac{l_q - \mathbf{c}^{*T} \boldsymbol{\xi}}{\sqrt{\mathbf{c}^{*T} \mathbf{c}^*}}\right)} \phi\left(\frac{l_q - \mathbf{c}^{*T} \boldsymbol{\xi}}{\sqrt{\mathbf{c}^{*T} \mathbf{c}^*}}\right).$$

3 Numerical illustration

In this section, we present a Monte Carlo study to assess the accuracy of the theoretical results derived in Lemma 2.1 and Theorem 2.2. The simulation study is conducted in two parts. In the first part, we examine the marginal densities of the components of the truncated normal vector under a linear constraint and compare the theoretical densities obtained from Lemma 2.1 with kernel density estimates based on simulated samples. In the second part, we investigate the accuracy of the theoretical expression for the mean vector given in Theorem 1 by comparing it with empirical averages computed from simulated truncated samples.

3.1 Assessment of Lemma 2.1

We first study the validity of Lemma 2.1 for the marginal distributions of the components of the truncated vector

$$\mathbf{Y} \stackrel{d}{=} \mathbf{Z} \mid (a < \mathbf{c}^\top \mathbf{Z} < b),$$

where $\mathbf{Z}$ is trivariate standard normal. In this part, we take

$$\mathbf{c} = (-0.4, 0.7, 0.2)^\top.$$

The truncation points a and b are chosen from quantiles of the linear combination $\mathbf{c}^\top \mathbf{Z}$ under two settings:

$$(q_1, q_2) = (0.30, 0.60), \quad (q_1, q_2) = (0.80, 0.95).$$

Thus, the first setting corresponds to a moderate central truncation interval, while the second one represents a more upper-tail and asymmetric truncation region.

For each truncation setting, we generated 10,000 observations from the corresponding truncated distribution and then estimated the marginal densities of Y_1 , Y_2 , and Y_3 by kernel smoothing using the `density` function in R. These empirical densities were then compared with the corresponding theoretical densities obtained from Lemma 2.1.

Figure 1 displays the results. The figure is arranged in a 3×2 layout: the rows correspond to the components Y_1 , Y_2 , and Y_3 , while the two columns correspond to the two choices of the truncation interval (a, b) . In each panel, the solid curve represents the theoretical density, whereas the dashed curve represents the kernel density estimate based on the simulated sample.

The agreement between the theoretical and empirical curves is remarkably close in all cases. This confirms that the marginal density formula given in Lemma 1 provides an accurate description of the univariate components of the multivariate normal distribution truncated by a linear constraint. The fit remains very

good both for the moderate truncation interval and for the more extreme upper-tail truncation case, which provides strong numerical support for the lemma.

3.2 Assessment of Theorem 2.2

We next evaluate the theoretical expression for the mean vector given in Theorem 2.2. In this part, we take the mean vector of the original multivariate normal distribution as

$$\boldsymbol{\xi} = (1, 1, 1)^\top,$$

and the coefficient vector in the linear constraint as

$$\mathbf{c} = (0.2, 0.3, 0.4)^\top.$$

We consider three covariance structures:

$$\Sigma_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \Sigma_2 = \begin{pmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 1 \end{pmatrix},$$

and

$$\Sigma_3 = \begin{pmatrix} 1.0 & 0.75 & 0.0 \\ 0.75 & 1.2 & -0.55 \\ 0.0 & -0.55 & 0.8 \end{pmatrix}.$$

For the truncation interval, we again use quantiles of the linear combination $\mathbf{c}^\top \mathbf{X}$. Two cases are considered:

$$\text{Case 1: } (q_1, q_2) = (0.20, 0.80), \quad \text{Case 2: } (q_1, q_2) = (0.40, 0.85).$$

The first case corresponds to a symmetric central truncation interval, while the second one corresponds to a more asymmetric interval. For each covariance structure and each truncation case, we computed the theoretical mean vector using the formula in Theorem 2.2 and compared it with the empirical mean obtained from 10^6 simulated observations from the corresponding truncated normal distribution.

The results are reported in Table 1. The numerical agreement between the theoretical and simulated values is excellent across all three components, all covariance structures, and both truncation settings. The differences are negligible and can be attributed to Monte Carlo error. These results provide clear numerical confirmation of Theorem 2.2 and show that the proposed expression for the mean vector of the linearly truncated multivariate normal distribution is highly accurate.

Overall, the simulation study strongly supports the theoretical developments of this paper. The density comparisons verify the marginal distributional result in Lemma 1, while the numerical agreement between

Table 1: Theoretical and simulated componentwise means of the truncated trivariate normal distribution under the linear constraint $a < \mathbf{c}^\top \mathbf{X} < b$, for $\boldsymbol{\xi} = (1, 1, 1)^\top$, $\mathbf{c} = (0.2, 0.3, 0.4)^\top$, and three covariance structures. Case 1 corresponds to the quantile pair $(0.20, 0.80)$, and Case 2 corresponds to $(0.40, 0.85)$. Simulated values are based on 10^6 generated observations.

Case	Component	Σ_1		Σ_2		Σ_3	
		Theo.	Sim.	Theo.	Sim.	Theo.	Sim.
Case 1	Y_1	0.4999	0.4996	0.2828	0.2821	-0.3238	-0.3249
	Y_2	0.2498	0.2501	0.2176	0.2179	0.0967	0.0959
	Y_3	-0.0003	-0.0004	0.1524	0.1519	0.5172	0.5182
Case 2	Y_1	0.5846	0.5831	0.4690	0.4693	-0.1299	-0.1304
	Y_2	0.3769	0.3771	0.4207	0.4207	0.2290	0.2292
	Y_3	0.1693	0.1697	0.3724	0.3725	0.5879	0.5887

theoretical and empirical mean vectors confirms the validity of Theorem 2.2. Hence, the proposed formulas provide an accurate and practically useful framework for analyzing multivariate normal risks under linear truncation constraints.

Discussion and Conclusion

In this paper, we have proposed a modified version of the MTCE measure, called MTCE-LC, which solve a limitation of MTCE measure by conditioning on a weighted sum of risk factors exceeding a high quantile rather than requiring simultaneous exceedances of all marginal quantiles. We have illustrated the accuracy of the results using a numerical illustration. It will, of course, be of interest to look for such measures for the class of elliptical, skew-elliptical and scale-mixtures of multivariate normal distributions and we are currently looking into this problem and hope to report the findings in a future paper.

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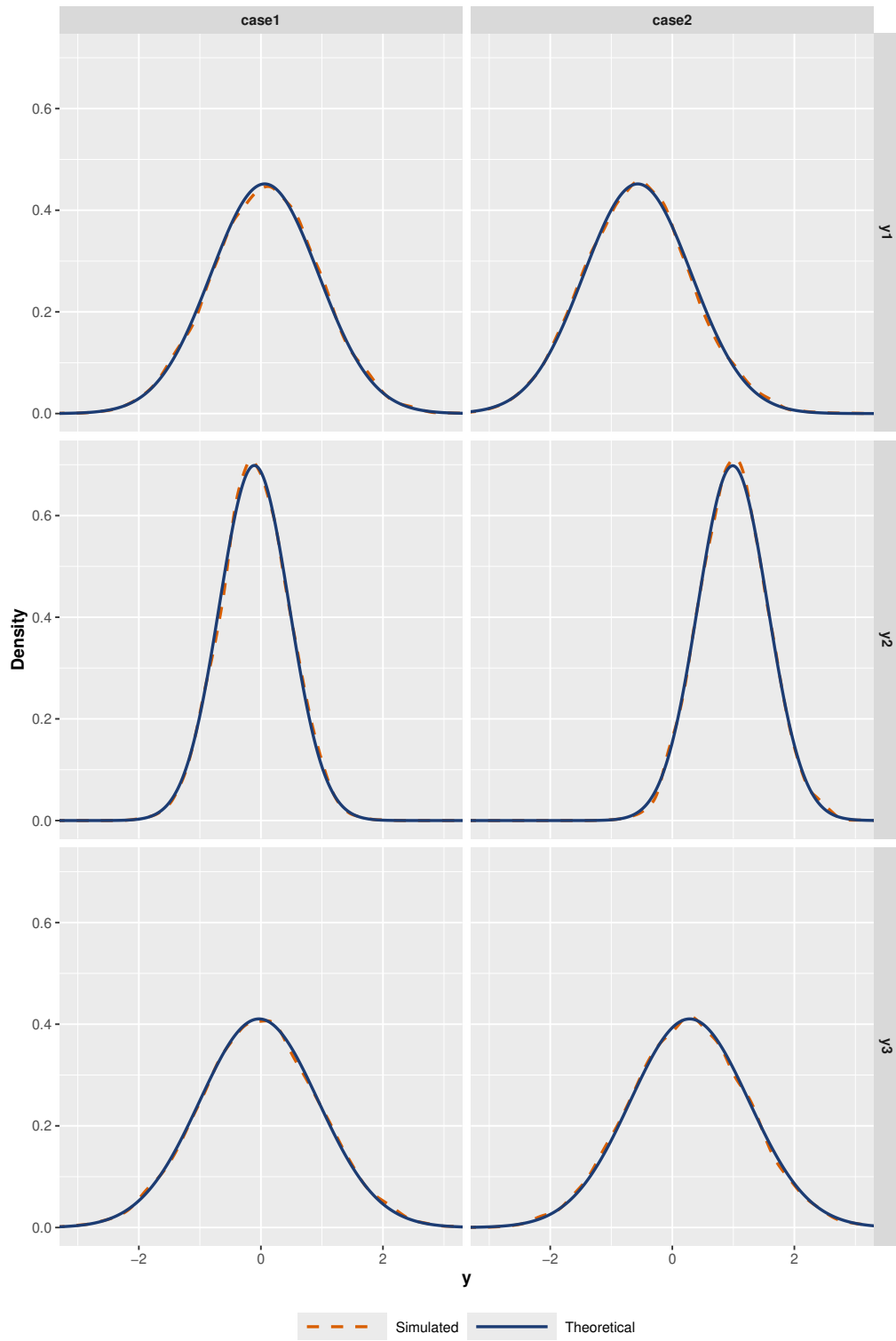


Figure 1: Comparison between the theoretical marginal densities from Lemma 1 and the kernel density estimates obtained from 10,000 simulated observations from the trivariate truncated normal distribution under a linear constraint. The rows correspond to Y_1 , Y_2 , and Y_3 , and the columns correspond to the two truncation settings determined by the quantile pairs $(0.30, 0.60)$ and $(0.80, 0.95)$ of $\mathbf{c}^\top \mathbf{X}$, with $\mathbf{c} = (-0.4, 0.7, 0.2)^\top$. In each panel, the solid line denotes the theoretical density and the dashed line denotes the simulation-based kernel density estimate.